

Rethinking Efficient Decoder Design: A Systematic Characterization of Where Decoder Computation Helps

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Abstract

Efficient decoders substantially reduce the cost of volumetric medical image segmentation by uniformly shrinking decoder channels and removing high-resolution stages, yet Dice and FLOPs alone do not explain how this simplification changes predictions: they cannot reveal where additional decoder capacity improves a prediction, where it degrades one, or whether these changes are spatially concentrated and predictable. We study this through a controlled comparison of a full and a compressed 3D UX-Net decoder that share an identical encoder, counting per voxel the positive transitions the full decoder produces (efficient wrong, full correct) and the negative transitions (efficient correct, full wrong). On 13-organ abdominal CT, the full decoder's net benefit is small and bidirectional—about 0.1% of voxels, largely offset by degradations—yet concentrated at organ boundaries and small structures and predictable from the efficient model's own uncertainty: at a predeclared budget, entropy-ranked contiguous 3D regions capture the net benefit at roughly five times the rate of matched-random regions, with a positive paired subject bootstrap. These findings show why aggregate accuracy and computation are insufficient for understanding efficient decoder design, and provide an evaluation protocol for region-adaptive decoder allocation rather than uniform compression.

1. Introduction

Volumetric medical image segmentation is increasingly driven by heavy U-Net-based architectures, particularly ones with transformer or large-kernel decoders that recover fine spatial detail through multi-scale fusion [4, 7, 11]. Their cost grows rapidly with resolution: the number of voxels grows cubically with the per-axis resolution, dense 3D convolution scales linearly in that voxel count, and global self-attention scales quadratically in it. The decoder, with its wide, high-resolution stages, often accounts for a large share of this cost, which limits deployment in real-

time and resource-constrained settings.

Efficient-decoder methods reduce this burden. Ef-fiDec3D [9] uniformly shrinks decoder channels and removes the highest-resolution stages, preserving aggregate accuracy across twelve tasks while cutting decoder parameters and FLOPs by an order of magnitude. Such methods answer how much decoder computation can be removed on average, but not where the removed computation would have helped. They apply the same reduction to every subject, organ, and spatial location, implicitly assuming that decoder capacity is useful roughly uniformly. Aggregate Dice and FLOPs cannot test this assumption: they reveal neither where additional decoder capacity improves a prediction, where it degrades one, nor whether such changes are spatially concentrated and predictable. A location that is hard to segment is not necessarily one a stronger decoder can fix, and a confident location may still be correctable.

This paper characterizes where the full decoder's computation actually changes predictions. We compare a full and an efficient decoder that share an identical encoder, so that every difference is attributable to decoder capacity, and we count, per voxel, the *positive transitions* the full decoder produces (efficient wrong, full correct) and the *negative transitions* it produces (efficient correct, full wrong). We then test whether the efficient model's own uncertainty identifies where the net benefit occurs, using subject-level statistics throughout. More specifically, our contributions are:

- **A transition-count comparison of efficient and full decoders.** Instead of comparing only aggregate Dice and FLOPs, we count per-voxel positive, negative, and net transitions with subject-level bootstrap, so that not every disagreement is counted as an improvement.
- **Evidence that decoder benefit is small, bidirectional, and spatially concentrated.** The net benefit is only about 0.1% of voxels and is largely offset by negative transitions, yet it is far from uniform: it concentrates at organ boundaries (boundary error $3.93\times$ interior) and small organs (size-difficulty $\rho = -0.54$; entropy predicts diffi-

075 culty beyond organ size, partial $r = 0.81$).
 076 • **Evidence that this benefit is predictable at test time.**
 077 At a predeclared 20% budget, ranking contiguous 3D re-
 078 gions by the efficient model’s entropy captures the full de-
 079 coder’s net benefit at about five times the rate of matched-
 080 random regions, with a positive paired subject-bootstrap
 081 lower bound—identifying where a stronger decoder helps
 082 without ground truth.

083 We deliberately stop short of proposing an adaptive
 084 method: whether these improvements can be realized as
 085 measured compute savings is the subject of a separate
 086 method study. Our goal here is to establish whether the em-
 087 pirical premise for such a method holds, and to provide an
 088 evaluation protocol for it.

089 2. Related Work

090 **Efficient decoders for 3D segmentation.** U-shaped
 091 encoder–decoder networks dominate volumetric segmenta-
 092 tion. Recent backbones raise accuracy with heavier compu-
 093 tation: UNETR [4] and Swin UNETR [5, 11] use trans-
 094 former encoders, while 3D UX-Net [7] and MedNeXt [10]
 095 use large-kernel convolutions, and nnU-Net [6] shows that
 096 pipeline configuration alone is strongly competitive. In
 097 several of these models the decoder, with its wide, high-
 098 resolution stages, dominates cost. EffiDec3D [9] uniformly
 099 reduces decoder channels and removes high-resolution
 100 stages, preserving aggregate accuracy across twelve tasks
 101 at much lower cost. These methods answer *how much* de-
 102 coder computation can be removed on average; we instead
 103 ask *where* the removed computation would have helped, by
 104 comparing a matched full and efficient decoder rather than
 105 proposing a new compression block.

106 **Uncertainty and difficulty estimation.** Predictive en-
 107 tropy, confidence, and Monte-Carlo dropout [2] are widely
 108 used to flag ambiguous predictions. Such signals identify
 109 where a model is uncertain, but ambiguity is not the same
 110 as decoder benefit: an uncertain voxel may remain wrong
 111 after additional decoding, and a confident voxel may still be
 112 correctable. We therefore evaluate signals against held-out
 113 *net transitions* between matched decoders, not against error
 114 alone.

115 **Conditional computation and adaptive inference.** Dy-
 116 namic networks allocate computation input-adaptively, spa-
 117 tially, or per sample [3]. Realizing such allocation for a 3D
 118 decoder presupposes that the useful computation is spatially
 119 concentrated and identifiable before it is executed—an em-
 120 pirical premise that is usually assumed rather than tested.
 121 Our study isolates and tests exactly this premise; designing
 122 a trained adaptive decoder is left to future work.

Evaluation and reproducibility. Small medical datasets
 create substantial seed and split variance, and voxel-level
 statistical tests can produce severe pseudo-replication. Our
 protocol reports seed-level results, discloses deviations
 from published baselines, and bootstraps subjects rather
 than voxels. This emphasis on auditing what aggregate met-
 rics conceal aligns with the goals of the WACV Evaluations
 & Datasets track.

3. Where the Full Decoder Helps

Our goal is to characterize *where* the extra computation in a
 full 3D decoder changes predictions relative to an efficient
 decoder, rather than to summarize it with a single accuracy
 or FLOPs number. This section defines the matched com-
 parison, the per-voxel prediction transitions we count, the
 test-time signal we evaluate, and the region-level opportu-
 nity analysis.

Matched decoder comparison. We compare two mod-
 els that share an identical encoder and differ only in the
 decoder. Full-UX uses the original wide, high-resolution
 decoder; Effi-UX uses the EffiDec3D decoder, which uni-
 formly reduces channel width and removes the highest-
 resolution stages [9]. Because the encoder is held fixed,
 any difference in the two models’ predictions is attributable
 to decoder capacity alone; the study therefore says nothing
 about the encoder, which we leave untouched. Let $\hat{y}_E(v)$
 and $\hat{y}_F(v)$ denote the efficient and full predicted labels at
 voxel v , and $y(v)$ the ground-truth label.

Prediction transitions. Rather than treating every dis-
 agreement between the two decoders as an improvement,
 we separate the two directions of change with plain indi-
 cator counts. A *positive transition* is a voxel the efficient
 decoder gets wrong and the full decoder gets right; a *nega-*
tive transition is the opposite:

$$P(v) = \mathbb{I}[\hat{y}_E(v) \neq y(v) \wedge \hat{y}_F(v) = y(v)], \quad (1)$$

$$N(v) = \mathbb{I}[\hat{y}_E(v) = y(v) \wedge \hat{y}_F(v) \neq y(v)], \quad (2)$$

$$U(v) = P(v) - N(v). \quad (3)$$

$U(v)$ is the *net* transition. The corresponding rates over a
 volume are $R_{\text{pos}} = \text{mean}_v P(v)$, $R_{\text{neg}} = \text{mean}_v N(v)$, and
 $R_{\text{net}} = R_{\text{pos}} - R_{\text{neg}}$. Reporting P , N , and U together
 prevents the common error of counting negative transitions
 as part of the decoder’s benefit.

Test-time signal. A benefit that could guide computation
 must be identifiable *without* the ground truth. We therefore
 evaluate whether the efficient model’s own uncertainty pre-
 dicts where net transitions occur. The primary signal is the

per-voxel predictive entropy of the efficient model,

$$H(v) = - \sum_{c=1}^C p_c(v) \log p_c(v), \quad (4)$$

where $p_c(v)$ is its softmax probability for class c and C is the number of classes. We compare H against confidence $\max_c p_c(v)$ and Monte-Carlo dropout [2] as alternative cheap signals.

Region-level opportunity analysis. Isolated voxels cannot be processed efficiently by standard 3D convolutions, so a deployable analysis must operate on contiguous regions. We partition the foreground-union volume into non-overlapping 16^3 blocks, rank blocks by mean entropy, and select the top fraction of blocks at a predeclared budget. Each selected block is expanded by a 4-voxel context halo, and we report the *executed* volume fraction this induces. Within the selected region we measure net transition utility, defined as $(\sum P - \sum N) / \sum P$ over the foreground union, and compare it against a matched-random selector that draws the same number of blocks. A voxel-level ranking of the same signal is reported only as a non-deployable oracle upper bound.

Statistics. All confidence intervals resample *subjects*, not voxels: with only twelve validation volumes, treating millions of correlated voxels as independent would grossly overstate significance. Correlations between H and U are computed per subject over entropy quantile bins and aggregated across subjects; the opportunity comparison uses a paired subject bootstrap in which the entropy and random selectors share the same resampled subject indices.

4. Experimental Protocol

Objectives. Using the matched comparison of Section 3, our analysis establishes three things left unresolved by aggregate efficient-decoder evaluation: how large and how bidirectional the full decoder’s benefit is, where it is concentrated anatomically, and whether the beneficial regions can be identified at test time from the efficient model’s uncertainty. Section 5 reports these in turn, preceded by the controlled-reimplementation baselines.

Dataset and split. The primary task is the 13-organ BTCV/Synapse abdominal CT benchmark with 18 training and 12 validation volumes. The evaluated structures are spleen, right and left kidney, gallbladder, esophagus, liver, stomach, aorta, inferior vena cava, portal and splenic veins, pancreas, and right and left adrenal glands. Images are intensity-windowed and resampled following the public EffiDec3D pipeline; training uses 96^3 patches. The same

validation subjects are paired across every model comparison.

Models and training. Full-UX is the public full 3D UX-Net [7] configuration with feature widths [48, 96, 192, 384]; Effi-UX replaces its decoder with the EffiDec3D decoder (48 channels, additive skip aggregation, resolution factor two). Both are trained for 45k iterations with a Dice–cross-entropy loss and AdamW. We evaluate one Full-UX run and two independently initialized Effi-UX runs. Sliding-window inference uses a 96^3 window, batch size four, overlap 0.7, and constant blending. We report Dice and 95th-percentile Hausdorff distance (HD95), together with parameters, multiply–accumulate operations (MACs), latency, and peak inference memory.

Reimplementation status. Our data were reconstructed from the public TransUNet/Synapse release, whose NIfTI files carry identity affine metadata, whereas EffiDec3D refers to a processed `btcv_trns` dataset that is not distributed. We therefore present the experiment as a *controlled reimplementation*, not an exact reproduction, and disclose the discrepancy from published numbers in full. Because every observation is relational—matched full versus efficient on identical local data—a residual reproduction gap for any one cell affects at most the estimated *magnitude* of that cell’s opportunity, which we bound with net (not positive-only) transitions and subject-level intervals; it does not change the direction of the finding. We verified that this gap is not a precision artifact: re-evaluating the same weights in FP32 and BF16 gives identical Dice.

Architecture generality. We organize architecture coverage as a predeclared ladder over the backbones for which EffiDec3D actually defines a decoder pair, rather than treating one extra model as proof of universality. The primary causal comparison is the matched 3D UX-Net pair (large-kernel CNN); a matched Swin UNETR [11] pair tests transfer to a hierarchical Transformer, with SwinUNETR-V2 [5] and MedNeXt [10] available as optional deeper pairs. We deliberately exclude UNETR [4] and nnU-Net [6]: EffiDec3D builds an efficient decoder only where a wide, high-resolution decoder bottleneck exists, and pairs one with neither (UNETR is already lightweight; nnU-Net self-configures its decoder). A matched transition comparison is therefore undefined for them, and an unofficial EffiDec3D variant would be an ungrounded extension rather than a control.

Dataset generality. The original EffiDec3D evaluation spans FeTA [8], BTCV, and the ten Medical Segmentation Decathlon (MSD) tasks [1]. We preselect three to

Table 1. BTCV controlled reimplementation. Published values are shown only for calibration; our paired analyses use the local runs. MAC denotes multiply-accumulate operations under our profiler.

Model	Dice $\uparrow$	HD95 $\downarrow$	MAC (G)	Params (M)	Lat. (ms)
Published Full-UX	.7974	—	631.97	53.00	—
Published Effi-UX	.7925	10.12	51.47	3.16	—
Full-UX, seed 0	.7918	9.04	578.74	53.007	40.6
Effi-UX, seed 0	.7700	14.41	41.06	2.955	7.5
Effi-UX, seed 1	.7755	10.17	41.06	2.955	7.6

four representative tasks: BTCV abdominal CT as the primary multi-organ task; FeTA fetal-brain MRI for modality and anatomy shift; MSD Task01 BrainTumour for multi-modal lesion segmentation; and either MSD Task06 Lung or Task08 HepaticVessel as a sparse-lesion or thin-structure stress test. Every dataset uses the same positive, negative, and net transition definitions, concentration analysis, opportunity curve, and subject-level intervals, and the set is frozen before inspecting results.

[TODO: Insert the matched Swin UNETR replication (opt. SwinUNETR-V2/MedNeXt) and the predeclared FeTA/MSD results; all currently completed numbers are 3D UX-Net on BTCV.]

5. Results

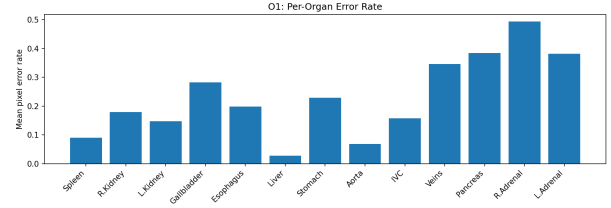
5.1. Controlled Reimplementation

Table 1 shows that the full model closely approaches the published Dice (a 0.56-point difference), whereas the two efficient runs remain 1.70–2.25 points below the published efficient model. The two-seed efficient mean is 0.7728, and seed 1 nearly matches the published HD95. The local Full-UX–Effi-UX Dice gap (1.63–2.18 points) is larger than the published 0.49-point gap; we do not attribute its full magnitude to decoder compression. Efficiency trends reproduce strongly: Effi-UX uses $14.1\times$ fewer MACs, $17.9\times$ fewer parameters, $5.3\text{--}5.4\times$ lower latency, and roughly $7.7\times$ lower peak inference memory than Full-UX.

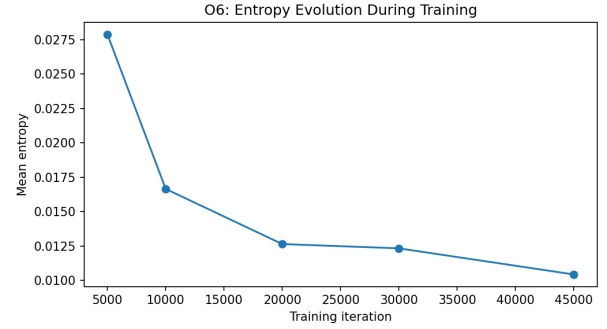
5.2. Errors and Uncertainty Are Anatomically Concentrated

The premise for selective allocation is that difficulty is not spatially uniform. Mean boundary error is 0.1892 versus 0.0481 in the eroded interior, a ratio of 3.93 (Fig. 1a). Small and thin structures are disproportionately hard: the organ-size/difficulty Spearman correlation is $\rho = -0.54$, and entropy retains strong association with organ difficulty after controlling for log volume (partial $r = 0.81$), so size alone does not explain the signal.

Uncertainty is likewise sparse: the median voxel entropy is near zero and only 1.18% of voxels exceed 0.5 (Fig. 2).



(a) Per-organ error.



(b) Entropy over training.

Figure 1. Anatomical and temporal structure of difficulty (Effi-UX, seed 1). Error concentrates at boundaries and small organs; entropy contracts during training but a residual difficult subset persists.

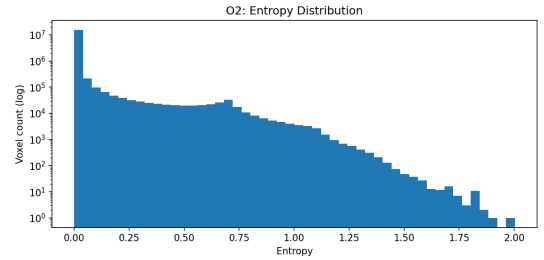


Figure 2. Voxel entropy distribution (Effi-UX, seed 1; log-count). Uncertainty is spatially sparse: the median voxel entropy is near zero and only 1.18% of voxels exceed 0.5.

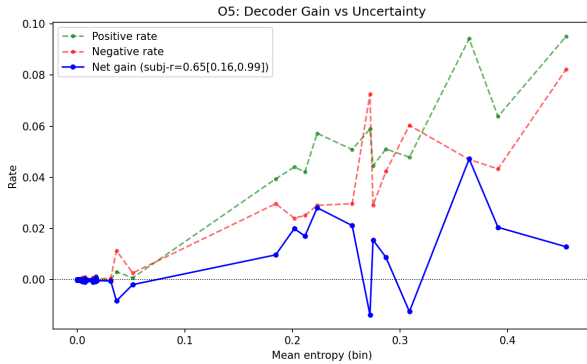
Difficulty also evolves systematically—mean entropy falls from 0.0279 at 5k iterations to 0.0166, 0.0126, 0.0123, and 0.0104 at 10k, 20k, 30k, and 45k (Fig. 1b)—but training removes global ambiguity without eliminating the concentrated residual. Most voxels are confidently easy; any benefit from a stronger decoder must come from the sparse remainder.

5.3. The Full Decoder’s Benefit Is Small and Bidi-rectional

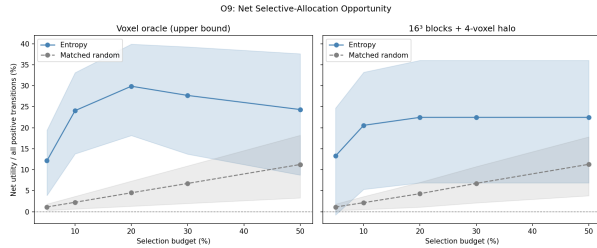
The full decoder is not uniformly better. For seed 0 it improves 0.339% of voxel predictions while degrading 0.209%; for seed 1 the values are 0.290% and 0.225% (Table 2). The net label-transition benefit is therefore only

Table 2. Transition-count observations across efficient-decoder seeds. Rates are fractions of evaluated voxels. The top-20% recovery is a voxel oracle (Sec. 3), not the deployable region result.

Metric	seed 0	seed 1
Positive transition rate	.00339	.00290
Negative transition rate	.00209	.00225
Net transition rate	.00130	.00065
Subject entropy–net r	.665	.646
Bootstrap 95% CI	[.171,.996]	[.158,.994]
Top-20% positive recovery	.864	.864



(a) Net transition vs. entropy.



(b) Region-level opportunity curve.

Figure 3. Predictability (seed 1). Net transitions rise with entropy (a); ranking contiguous 16^3 blocks by entropy recovers net transition utility far above a matched-random selector (b).

0.065–0.130% of voxels, largely offset by negative transitions. Although this label-level statistic is not identical to Dice, it shows why positive and negative transitions must be reported together: reporting positive transitions alone would overstate the decoder’s benefit roughly twofold.

5.4. The Benefit Is Spatially Predictable

The subject-level entropy–net-transition correlation is positive for both seeds, with a subject-bootstrap lower bound above zero (Table 2, Fig. 3a). We convert this into a deployable test with the contiguous-region selector of Section 3. At the predeclared 20% block budget—which executes 29.1% of the foreground-union volume after halo

expansion—entropy-ranked regions attain a net utility of 0.225 versus 0.043 for random selection, a roughly five-fold gain, with a paired 95% lower bound of $0.056 > 0$ (Fig. 3b). The advantage is significant at every budget of 10% or more and vanishes only at 5%. Crucially, the selected regions capture all positive transitions but also 77.5% of the negative ones, so the honest headline is the *net* gain, not positive recovery. A voxel-level oracle that ranks individual voxels and counts only positive transitions places 86.4% of them in the top 20% (Table 2); we report it solely as a non-deployable upper bound.

Finally, we compare cheap routing signals against held-out net transitions. Confidence and entropy behave almost identically ($r = 0.663$ and 0.655); Monte-Carlo dropout is not a valid comparator for this checkpoint because the architecture has no active dropout, so its stochastic variance is effectively zero. We therefore retain confidence as a required baseline and do not claim entropy is uniquely optimal.

[TODO: Add the matched Swin UNETR replication and the predeclared FeTA/MSD cells; current results are 3D UX-Net on BTCV only.]

6. Discussion and Limitations

What the evidence supports. Across two efficient-decoder seeds, the direction of the central observation is stable: the full decoder’s aggregate benefit is small and bidirectional, and its net benefit rises with the efficient model’s uncertainty. Boundary and organ analyses show that difficulty is spatially heterogeneous, and the region-level opportunity curve shows that entropy-selected contiguous regions carry disproportionate net benefit. Together these justify testing selective regional decoding rather than a uniformly expensive decoder.

What it does not yet support. The study does not demonstrate lower *executed* compute for an adaptive model. The opportunity curve uses full-model predictions to define benefit retrospectively and does not account for errors introduced by a learned refinement module; the voxel-level 86.4% recovery figure is an oracle upper bound, not a realizable router. Dynamic FLOPs, latency, memory, and final Dice must be evaluated in a separate method study. The analysis is also *decoder-only* by construction: because EffiDec3D alters the decoder while holding the encoder identical, every transition isolates decoder capacity, but the study says nothing about the encoder, which accounts for the larger share of compute. Whether encoder computation is similarly over-provisioned is a distinct question requiring representational probes rather than a transition pair, and is left to future work.

Reproduction gap. Our efficient baseline remains 1.70–2.25 Dice points below the published result, while the full baseline is much closer. The most plausible unresolved difference is preprocessing: our reconstructed files have identity affine metadata, whereas the original work used a processed dataset not distributed with the code. We tested the obvious remedy—disabling spatial resampling—and it was decisively worse (≈ 10 Dice points), so bypassing resampling is not the fix. The enlarged local full-efficient gap could inflate the apparent amount of decoder opportunity; we mitigate this by disclosing all values, analyzing two seeds, and limiting claims to controlled local comparisons and net (not positive-only) transitions.

Statistical limitations. The validation set contains only twelve subjects, so confidence intervals are wide and millions of voxels must not be treated as independent. The entropy-error coefficient we report ($r = 0.97$) pools bins across subjects and is retained only as a descriptive diagnostic; final error predictability should use subject-level AUROC/AUPRC and calibration.

Generalization. All completed results use one CT dataset and one backbone. Without cross-backbone and cross-modality replication, the finding may be specific to 3D UX-Net, the Synapse split, or EffiDec3D’s resolution restriction. The planned ladder uses matched pairs across the EffiDec3D backbone family (3D UX-Net and Swin UNETR, optionally SwinUNETR-V2 and MedNeXt) and representative tasks from the original EffiDec3D scope; we restrict it to backbones EffiDec3D actually equips with an efficient decoder, since a matched transition comparison is undefined otherwise. The final claim will be narrowed if these experiments do not reproduce the direction.

Clinical interpretation. High uncertainty and small-organ difficulty do not imply clinical safety. Entropy can miss confidently wrong predictions, and improving average Dice does not guarantee preservation of clinically important boundaries. Any future adaptive decoder should retain a conservative fallback and be evaluated per organ and per subject, not only by average compute.

7. Conclusion

We presented a transition-count perspective on efficient 3D segmentation decoders. A controlled full-versus-efficient comparison, holding the encoder fixed, shows that additional decoder capacity has a small and bidirectional aggregate effect, that its remaining benefit is concentrated at boundaries and small structures, and that this benefit is predictable at test time: entropy- ranked contiguous regions recover the net benefit far above matched-random selection

at a predeclared budget, with a positive paired lower bound. These trends are stable across two efficient-decoder seeds despite an unresolved absolute reproduction gap.

More broadly, the results show why aggregate Dice and FLOPs are insufficient for understanding efficient decoder design: error, uncertainty, and decoder-specific benefit are distinct quantities, and only their separation reveals that efficient decoders are over-provisioned *non-uniformly*. The net-transition, paired-bootstrap, and contiguous-region protocol provides a stricter test of whether selective allocation is empirically justified, and an evaluation template that other efficient-decoder studies can reuse.

We deliberately stop short of proposing an adaptive method: whether the observed opportunity can be realized as measured compute savings, and whether the untouched encoder is similarly over-provisioned, are left to future work.

[TODO: Once the matched Swin UNETR and pre-declared FeTA/MSD cells are in, state the cross-backbone/cross-dataset outcome here; if the direction fails to replicate, narrow “predictable” to the tested setting rather than dropping it.]

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